

Title: Phishing Emails Investigation Report

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CASE 0

Executive Summary

I responded to an alert about a potential phishing email targeted at faculty members of the educational institute. The email appeared to come from a trusted entity who claims to be a Magister in Education and LEB Teacher at FESAD. After thorough analysis and deep investigation, it turned out to be a true positive.

Manual and Automated Header Analysis

From: ERIKA JOHANA LOPEZ VALIENTE <erikajohana.lopez@uptc.edu.co>

Date: Thu, 9 Dec 2022 09:58:26 +0100

Subject: COMMERCIAL PURCHASE RECEIPT ONLINE 27 NOV

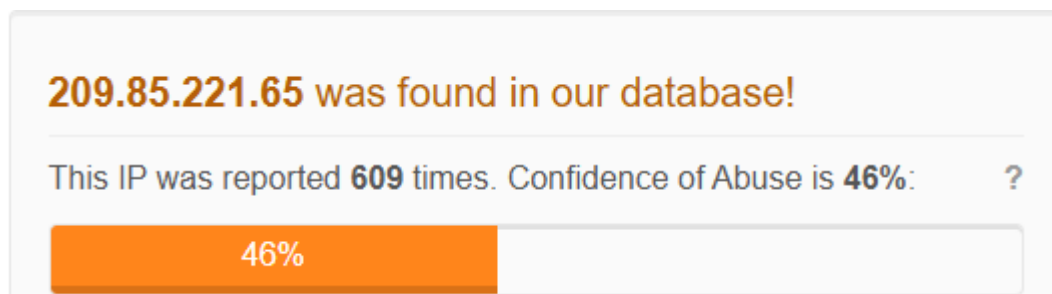
To: undisclosed-recipients;;

X-TM-Authentication-Results: spf=pass (sender IP address: 209.85.221.65)

Return-Path: erikajohana.lopez@uptc.edu.co

Red Flags

- Sender IP address has been reported 600+ times according to AbuseIPBD.



- The absence of a value in the **To** field confirms that the email was sent to multiple recipients.

- Even though it passed the spf check, analyzing the headers with MX toolbox shows that the email domain - *uptc.edu.co* is not DMARC compliant. Cross checking this fact with PhishTool suggests the same.

Header Analyzed

Email Subject: [COMMERCIAL PURCHASE RECEIPT ONLINE 27 NOV](#)

Copy/Paste Warning

Copy/Pasting a header works for most people, but sometimes it can cause problems with this

Delivery Information

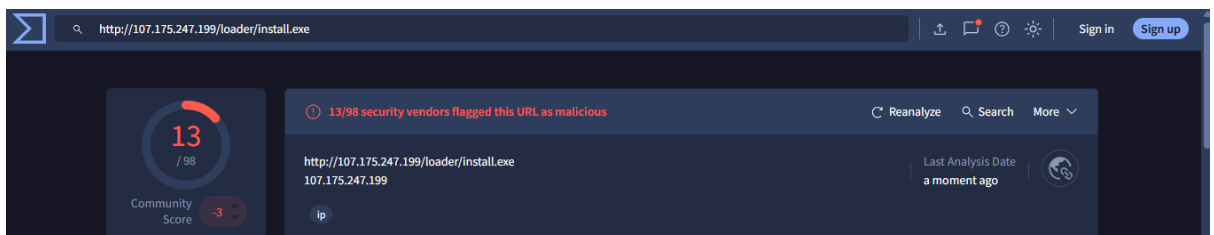
- ✖ DMARC Compliant
 - ✔ SPF Alignment
 - ✖ SPF Authenticated
 - ✔ DKIM Alignment
 - ✖ DKIM Authenticated

Body Analysis

- There is use of fear tactics.
- The attachment of a link to download an invoice instead of the standard practice of attaching a file.
- The link shows that an exe file will be downloaded instead of text files like pdf, xlsx etc.

Attachment Analysis

- Analyzing the attached link on Virus Total confirms that it is a malicious link which has been flagged by anti-virus vendors. PhishTool report also confirmed this.



- Analyzing further on VirusTotal shows that the IP address - *107.175.247.199* from the received header fields is linked to domain - [ripley.studio](#) which has been

repeatedly reported as malicious. The domain communicates files like *install.exe*.

Indicators of Compromise

Type	Indicator
Domain	uptc.edu.co
IP	107.175.247.199
Sender IP	209.85.221.65
URL	http://107.175.247.199/loader/install.exe

Risk Assessment

The email poses a high risk due to the multiple indicators that suggests that the sender is a bad actor.

Recommendations

- Listed IOCs should be blocked on mail gateways and firewall.
- Sender domain and email address should be reported as phishing and added to spam list immediately.
- Phishing awareness exercise should be organized for Staff.
- Faculty members should be notified, credentials should be reset and a deep virus scan should be carried out on their devices.

Conclusion

The phishing alert was a true positive as confirmed by the multiple indicators of compromise. The email should be flagged as a phishing mail and response actions such as containment, eradication and recovery should be taken with immediate effect.

Executive Summary

While monitoring the institution's email security dashboard, I observed an email sent to a fellow employee which has a link attachment that prompts the recipient to redeem a free voucher.

Log Analysis

From: free@coffeeshoop.com

To: Felix@letsdefend.io

Subject: Free Coffee Voucher

Date: May, 13, 2024, 09:22AM

Action: Allowed

Red Flags

Looking up the record of the domain – *coffeeshoop.com* with two different services shows that it has never been registered before. This reveals that the domain was spoofed and that it might be a case of clone phishing.

Whois Record for Coffeeshoop.com

Domain Available



coffeeshoop.com is for sale!
The domain you are researching is available for registration.

Buy coffeeshoop.com

Domain Profile

Domain Status Never Registered Before

Whois Record



Whois Record Not Available
This domain is not registered.

Body Analysis

- There is an element of urgency in the body of the email. The recipient is being urged to click the attached link.

- The attached image looks broken and this can signify a number of suspicious happenings including blockage by email client, imperfect mimicking of a legitimate template and the image might be hosted on a suspicious domain amongst other reasons.

Attachment Analysis

- VirusTotal analysis reveals that the URL has been flagged as malicious by a few vendors.
- In the **Relations** tab, Sophos classified it as malware and spyware.

alphaMountain.ai	Malicious (alphaMountain.ai)
Sophos	spyware and malware
Forcepoint ThreatSeeker	malicious web sites

- The value of the **Final URL** reveals a hash of the file - 59cbd215-76ea-434d-93ca-4d6aec3bac98. Analyzing the hash on VirusTotal showed no suspicion neither was it flagged by any vendor. However, analyzing it with Hybrid Analysis confirmed that the link does download a Trojan malware.

Analysis OverviewRequest Report DeletionSample (30KiB)Show Sample Content

Submission name:

Size:

Type:

Mime:

SHA256:

Submitted At:

Last Anti-Virus Scan:

Last Sandbox Report:

59cbd215-76ea-434d-93ca-4d6aec3bac98-free-coffee.zip

30KiB

data

compressed

zip

application/zip

6f33ae4bf134c49faa14517a275c039ca1818b24fc2304649869e399ab2fb389

2024-07-04 13:19:58 (UTC)

2025-05-30 09:59:58 (UTC)

2024-12-19 07:03:48 (UTC)

malicious

Threat Score: 100/100

AV Detection: 12%

Labeled As: Trojan.Generic

#urlscanio

#file

X Post

Link

E-Mail

Indicators of Compromise

Type	Indicator
Domain	coffeeshoop.com
File Hash	59cbd215-76ea-434d-93ca-4d6aec3bac98
URL	https://download.cyberlearn.academy/download/download?url=https://files-ld.s3.us-east-2.amazonaws.com/59cbd215-76ea-434d-93ca-4d6aec3bac98-free-coffee.zip

Risk Assessment

The email poses a high risk due to the multiple indicators that suggests that the sender is a bad actor with the most threatening indicator being the attachment of a trojan file.

Recommendations

- Listed IOCs should be blocked on mail gateways and firewall.
- Sender domain and email address should be reported as phishing and added to spam list immediately.
- Phishing awareness exercise should be organized for Letsdefend's Employees.
- Felix's device should be immediately contained.
- It should be confirmed if the email was sent to multiple employees.
- Credentials should be reset.
- Further investigations should be made including but not limited to – finding out if the file was opened, if it was, malware analysis and forensics should be completed to observe the behaviour of the malware.

Conclusion

The incoming mail was a phishing attempt as confirmed by the multiple indicators of compromise. The recommendations should be acted on and response actions such as containment, eradication and recovery should be taken with immediate effect.